Allocating Held-Out Time-Reversal Scores to Inversion Distinctions in Bach Chorales and the Weimar Jazz Database

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Abstract

We estimate forward-versus-reversed held-out log scores for transposition-quotiented harmonic transitions in 430 Bach chorales (music21 corpus, 26249 chord changes, 67 key-relative states) and 412 Weimar Jazz Database solos grouped into 273 compositions (27491 chord changes, 175 states). In the bilateral-support analysis at $r = 1$, retained-total and conditional scores across $\alpha \in \{0.1, 0.5, 1\}$ range from 0.765 to 0.832 bits per transition in Bach and from 1.555 to 2.670 bits per transition in WJazzD. A standard KL chain-rule identity gives both a nonnegative population decomposition and an exact eventwise decomposition of held-out log scores under an inversion quotient. For the inversion allocation, the event-weighted residual is 0.137 to 0.156 bits in Bach, or 16.0 to 16.2% of the event-weighted score; lower endpoints of two-sided 95% normal intervals for the equal-chorale mean are 0.128 to 0.145 bits. In WJazzD the corresponding values are 0.091 to 0.151 bits, 3.7 to 4.7%, and 0.053 to 0.089 bits for the equal-composition interval lower endpoints, all below the pre-specified 0.10-bit criterion. Three exploratory controls qualify the share contrast: at $\alpha = 0.5$, a three-stratum shared-support standardisation reduces the gap by 30.5%; across all α, fold-local complete-fiber alphabet restrictions leave a 11.3 to 21.0 percentage-point gap among retained events; and common-mode weighting leaves 5.9 to 7.1 points, while major-only estimates reverse the ordering. At $\alpha = 0.5$, stationary circulation reconstructions close to machine precision (reported error 0.0); named-window counts show that one reported high-affinity cycle has one observed occurrence. This applies irreversibility diagnostics to harmonic transition laws, whereas González-Espinoza, Martínez-Mekler and Lacasa (2020) studied melodic visibility statistics; no theorem novelty is claimed.

I. Logic

1 Prior musical irreversibility and claim boundary

Time irreversibility of music has been quantified before: González-Espinoza, Martínez-Mekler and Lacasa (2020) applied horizontal-visibility-graph statistics to scalar melodic note streams across 8,856 pieces. We quantify a different object, the reversal asymmetry of transposition-quotiented harmonic transition laws, and decompose it through inversion-sensitive, directed-edge and closed-cycle analyses. The quotient identity we use is a corollary of the KL chain rule and of exact coarse-graining results (Esposito, 2012; Teza and Stella, 2020); it is stated as a lemma because it fixes the estimand and prevents an incorrect quotient implementation, not as a contribution.

2 Harmonic edge laws, reversal and bilateral-support estimands

Definition 1. *A chord event is a key-relative 12-bit pitch-class mask with mode (the annotated tonic at pitch class 0); consecutive identical masks are collapsed to chord changes (an uncollapsed control is reported). A directed edge is a pair (i, j) of consecutive states; $R(i, j) = (j, i)$ is reversal. For an edge law μ, $\sigma(\mu) = D_{\mathrm{KL},2}(\mu \| R_*\mu)$ in bits.*

Held-out scoring: for work-grouped folds, each held-out edge receives $d_{ij} = \log_2 \frac{c_{ij}+\alpha}{c_{ji}+\alpha}$ from training counts; the event mean $\bar{s}$ is a predictive log score, not σ of a law. Bilateral support restricts to edges with $c_{ij}, c_{ji} \geq r$ in training; the retained-total estimand assigns excluded edges zero, the conditional estimand averages over retained edges only.

Smoothing and support convention. In each fold the fitted support E_f is the set of training edges together with their reversals. The fitted C_{12} law is $\widehat{P}_f(e) = (c_f(e) + \alpha)/Z_f$ for $e \in E_f$ with $Z_f = \sum_{e \in E_f}(c_f(e) + \alpha)$, and α/Z_f for a held-out edge outside E_f. The D_{12} law is obtained only by pushforward: $h_*\widehat{P}_f(z) = \sum_{e \in E_f,\, h(e)=z} \widehat{P}_f(e)$, and a held-out edge whose fiber meets E_f nowhere receives $\widehat{P}_f(e) + \widehat{P}_f(\kappa e)$. Pseudocounts are therefore never added per fiber cell. The κ-support of a held-out edge e is the indicator that $\kappa e \in E_f$ and $\kappa e \neq e$.

Folds and inference. Groups (chorales in Bach, compositions in WJazzD) are sorted by the SHA-256 of their identifier and assigned round-robin to five folds, so performances of one composition never cross folds. Corpus scores weight held-out events equally. Group inference first averages the score within each group and then weights groups equally; the pre-specified interval is the group mean minus 1.96 standard errors (the lower endpoint of a two-sided 95% normal interval, Table 2); the exploratory controls report the group mean minus 1.645 standard errors (a one-sided 95% lower bound, Tables 3 and 4). Because the weightings differ, a group bound need not lie below the event-weighted corpus score. A group with no retained held-out event under an alphabet restriction is excluded from the group mean and counted in the table.

3 Diagonal C_{12} quotient

Let $G = \mathbb{Z}_{12}$ act diagonally on edges, $T_g(i, j) = (gi, gj)$, and let reversal commute with the action. For the symmetrised law $\mu^G = \frac{1}{12}\sum_g (T_g)_*\mu$ and the orbit map q, $\sigma(\mu^G) = \sigma(q_*\mu)$ (KL chain rule for the statistic q; both symmetrised laws are uniform within each orbit, so the conditional term vanishes). Our key-relative representation is the tonic-anchored orbit representative; an exact unsmoothed check on the common support gives identity gap 0.0 in Bach. Smoothed plug-ins differ because pseudocounts must be pushed forward, not added per cell.

4 Exact $C_{12} \to D_{12}$ allocation

Lemma 1 (standard KL chain-rule corollary)**.** *Let h be a quotient commuting with reversal, P an edge law and $Q = R_*P$. Then*

$$D_{\mathrm{KL}}(P \| Q) = D_{\mathrm{KL}}(h_*P \| h_*Q) + \sum_z (h_*P)(z)\, D_{\mathrm{KL}}\big(P(\cdot \mid z) \| Q(\cdot \mid z)\big).$$

The second population term is nonnegative. For a fold-fitted predictive pair $(\widehat{P}_f, \widehat{Q}_f)$ the corresponding held-out identity is eventwise,

$$\log_2 \frac{\widehat{P}_f(x)}{\widehat{Q}_f(x)} = \log_2 \frac{h_*\widehat{P}_f(hx)}{h_*\widehat{Q}_f(hx)} + \log_2 \frac{\widehat{P}_f(x \mid hx)}{\widehat{Q}_f(x \mid hx)},$$

and its held-out residual is not guaranteed nonnegative by the theorem; positivity and its group bounds are empirical results.

Here κ is mode-conditioned pitch-class inversion ($p \mapsto -p$, mode fixed) and h identifies an edge with its inversion. We write $\bar{s}_{C_{12}} = \bar{s}_{D_{12}} + \bar{\delta}_{\mathrm{inv}}$ for the held-out event means. Under the stated estimator and mode-fixed action, $\bar{\delta}_{\mathrm{inv}}$ is the held-out residual allocated to distinctions within h-fibers; an event contribution can be nonzero only if its inverted edge has positive training mass. The mode-fixed action is one valid D_{12} action, not the unique music-theoretic one.

5 Detailed balance, cycle affinities and circulation

For a stationary first-order chain with flow $F_{ij} = \pi_i P_{ij}$, $\sigma = \sum_{i<j}(F_{ij} - F_{ji}) \log_2(F_{ij}/F_{ji})$, and $\sigma = 0$ iff detailed balance iff every cycle affinity $A_C = \sum_{(i,j)\in C} \log_2(F_{ij}/F_{ji})$ vanishes (Kolmogorov, 1936). With a spanning forest, $\sigma = \sum_C j_C A_C$ over fundamental cycles; individual terms are basis dependent, the total is not. A loop's affinity is a force, not a contribution: it is reported together with its circulation and with contiguous window counts.

II. Experiment

6 Corpora, states, grouped folds and pre-specification

Bach: 430 chorales, 26249 collapsed chord changes, 67 states; folds by chorale. WJazzD: 412 solos in 273 compositions, 27491 collapsed chord changes, 175 states; folds by composition. Chord types are triads (Bach, music21 quality) and thirteen jazz symbol templates (WJazzD). State alphabets and annotation pipelines differ, so all cross-corpus differences reported below are descriptive and conditional on these representations. The conjunctive decision rule required the equal-group mean residual minus 1.96 standard errors, the lower endpoint of a two-sided 95% normal interval, to exceed 0.10 bits in both corpora at every $\alpha \in \{0.1, 0.5, 1\}$; it also required closure error below 10^{-12}. The rule was fixed in the dated analysis plan of 2 September 2026 before the inversion allocation was computed; it was recorded in the project ledger, not deposited in a public registry, so we call it pre-specified rather than preregistered. The vocabulary, alphabet, mode, bootstrap and cycle analyses are exploratory.

7 Support-controlled held-out irreversibility

Endpoint (piece boundary) marginals contribute 0.2% (Bach) and 0.0% (WJazzD) of the plug-in σ; merging states with fewer than five occurrences changes nothing. Figure 1 shows the bilateral-support curves.

Table 1: Held-out reversal asymmetry by bilateral support (bits per transition). rho = share of held-out events on edges observed in both directions in training; retained-total assigns excluded edges zero; LB = work- (Bach) or composition- (WJazzD) level 95% lower bound.

α	support	ρ (%)	retained-total	conditional	LB	works with no retained edge
Bach chorales						
0.1	all	100.0	0.966	0.966	0.975	0
0.1	r=1	95.9	0.798	0.832	0.809	0
0.1	r=2	93.9	0.741	0.789	0.747	0
0.1	r=5	89.1	0.617	0.692	0.618	0
0.5	all	100.0	0.894	0.894	0.904	0
0.5	r=1	95.9	0.782	0.815	0.792	0
0.5	r=2	93.9	0.730	0.778	0.736	0
0.5	r=5	89.1	0.611	0.686	0.613	0
1	all	100.0	0.855	0.855	0.865	0
1	r=1	95.9	0.765	0.798	0.775	0
1	r=2	93.9	0.718	0.765	0.724	0
1	r=5	89.1	0.605	0.678	0.606	0
WJazzD						
0.1	all	100.0	3.209	3.209	2.700	0
0.1	r=1	60.9	1.625	2.670	1.411	14
0.1	r=2	56.3	1.486	2.641	1.297	17
0.1	r=5	46.5	1.137	2.445	0.962	20
0.5	all	100.0	2.715	2.715	2.304	0
0.5	r=1	60.9	1.591	2.614	1.382	14
0.5	r=2	56.3	1.463	2.600	1.276	17
0.5	r=5	46.5	1.129	2.426	0.954	20
1	all	100.0	2.490	2.490	2.120	0
1	r=1	60.9	1.555	2.555	1.350	14
1	r=2	56.3	1.438	2.556	1.254	17
1	r=5	46.5	1.118	2.404	0.945	20

8 Inversion allocation and group inference

The pre-specified conjunctive rule fails: the minimum interval lower endpoint is 0.053 (WJazzD, $\alpha = 1$). Bach exceeds the 0.10-bit interval-lower-endpoint criterion at every α for collapsed events; its collapsed-event share is 16.0 to 16.2%, and its uncollapsed share is 16.0% at the only tested value, $\alpha = 0.5$. WJazzD's collapsed share is 3.7 to 4.7%, and its uncollapsed share is 4.0% at $\alpha = 0.5$. The absolute residual ranges overlap (0.137 to 0.156 versus 0.091 to 0.151 bits), while WJazzD's inversion-invariant score is 3.3 to 3.8 times Bach's; both the residual numerator and the score denominator determine the reported shares. A cluster bootstrap that resamples chorales or compositions with replacement (2,000 draws, held-out scores fixed, no refit) gives 5th percentiles of the event-weighted residual of 0.121 to 0.139 bits in Bach and 0.073 to 0.119 bits in WJazzD, and 90% percentile bands for the share of 14.4 to 17.7% and 2.9 to 5.8% respectively (Table 2); the two share bands do not overlap at any α.

9 Controls on the share contrast: vocabulary, alphabet and mode

At $\alpha = 0.5$, standardising to the 3 inversion-closed chord-family strata shared by both corpora changes the inversion shares from 16.0% and 4.0% to 15.9% and 7.6%, reducing the gap from 0.120

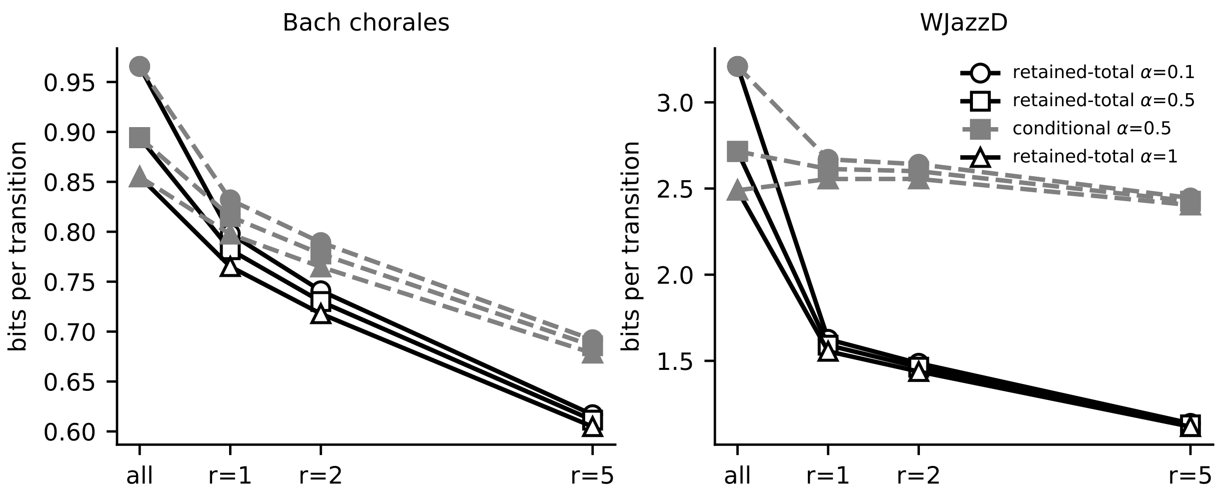

Figure 1: Bilateral-support control: retained-total (solid) and conditional (dashed) scores across support thresholds and smoothing.

Table 2: Exact eventwise allocation of held-out reversal log scores (bits per transition). Per-event closure error $\leq 9 \times 10^{-16}$. The barred scores, residual and share are event-weighted. The group mean is unweighted over chorales or compositions, and "95% CI lower" equals the group mean minus 1.96 standard errors, the pre-specified lower endpoint of a two-sided 95% normal interval. The last two columns are from a cluster bootstrap over groups (2,000 draws, no refit): the 5th percentile of the event-weighted residual and the 5th to 95th percentile band of the share.

corpus	α	$\bar{s}_{C_{12}}$	$\bar{s}_{D_{12}}$	$\bar{\delta}_{\text{inv}}$	share (%)	group mean	95% CI lower	boot. $\bar{\delta}$ p05	boot. share (%)
Bach	0.1	0.966	0.809	0.156	16.2	0.166	0.145	0.139	14.6–17.7
Bach	0.5	0.894	0.752	0.143	16.0	0.152	0.133	0.127	14.4–17.4
Bach	1	0.855	0.719	0.137	16.0	0.146	0.128	0.121	14.4–17.4
WJazzD	0.1	3.209	3.058	0.151	4.7	0.120	0.089	0.119	3.8–5.8
WJazzD	0.5	2.715	2.607	0.108	4.0	0.085	0.063	0.087	3.2–4.9
WJazzD	1	2.490	2.399	0.091	3.7	0.071	0.053	0.073	2.9–4.4
Bach (uncollapsed)	0.5	0.667	0.561	0.106	16.0	0.117	0.102	–	–
WJazzD (uncollapsed)	0.5	2.694	2.587	0.108	4.0	0.084	0.063	–	–

to 0.083 ($E = 0.30$). This restricted shared-support standardisation therefore does not remove most of the observed contrast, but it excludes the 29 jazz-only strata (two of which hold 48% of WJazzD events) and does not establish a vocabulary-independent corpus effect.

Across the 9 α-by-k settings, retained-event shares are 16.0 to 26.0% in Bach and 3.9 to 5.8% in WJazzD, yielding positive gaps of 11.3 to 21.0 percentage points. Achieved alphabet sizes differ at every target, and retained mass ranges from 64.6 to 99.7% in Bach and 46.3 to 75.0% in WJazzD, with 7 to 22 empty WJazzD groups. This control shows that the gap does not disappear under these fold-local retained-event restrictions; it does not establish alphabet independence.

Across α, common-mode weighting reduces observed gaps of 11.48 to 12.30 percentage points to 5.93 to 7.08 points, while major-only gaps are -3.42 to -2.77 points. This is attenuation under a fixed representation, not evidence that mode mixture causally accounts for a specified fraction of a corpus effect. The κ-support column identifies a necessary estimator-specific support gate under the mode-fixed action: support is 6.8% in Bach major and 84.0% in Bach minor (8.3% and 10.2% in WJazzD). It does not by itself explain residual magnitude, because contributions among

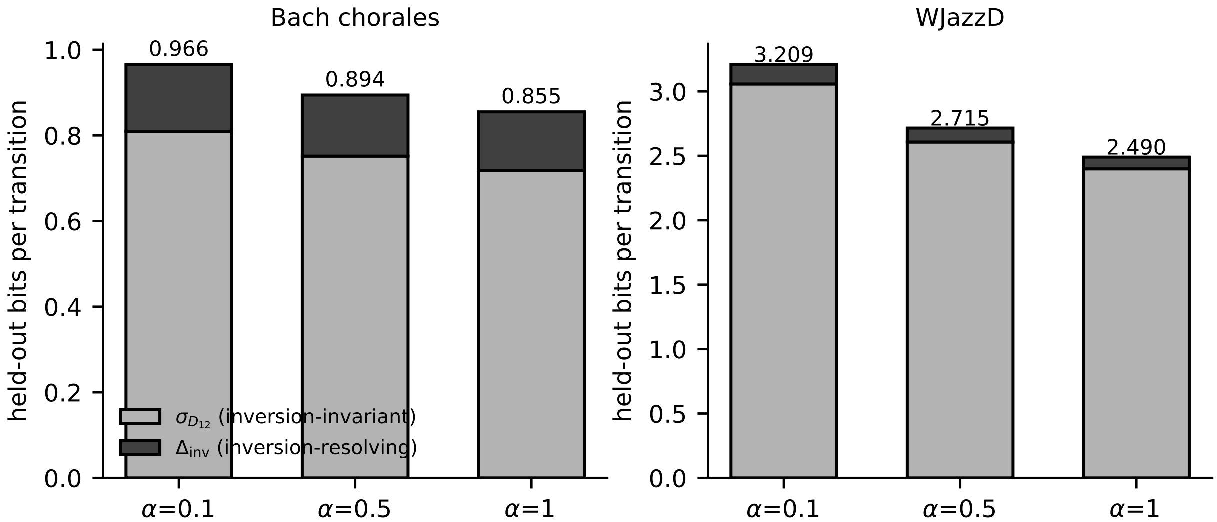

Figure 2: Exact allocation of held-out reversal scores into inversion-invariant and inversion-resolving parts.

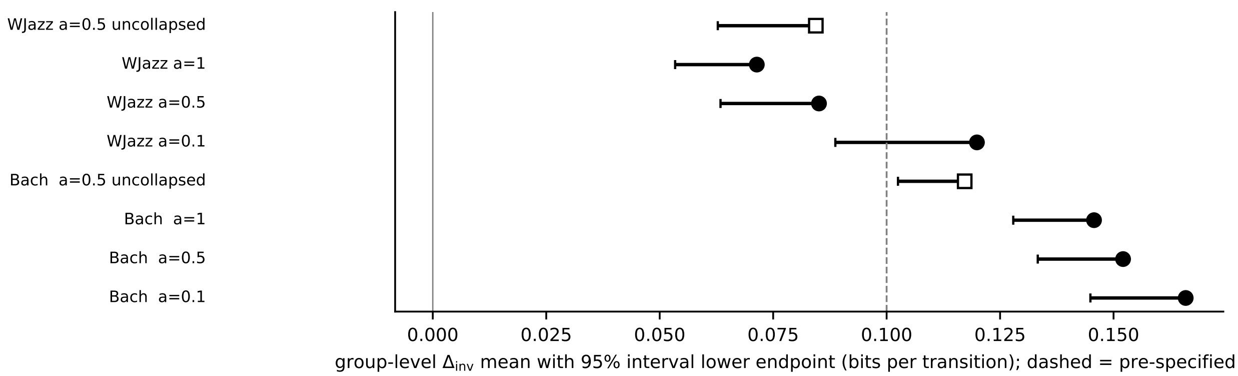

Figure 3: Group-level $\bar{\delta}_{\text{inv}}$ with 95% interval lower endpoints; dashed = pre-specified 0.10 bits.

supported events also differ (at $\alpha = 0.5$ the mean residual per supported event is 0.015 bits in Bach major, 0.385 in Bach minor, 1.052 in WJazzD major and 2.139 in WJazzD minor). Taken together, the gap persists under the limited shared-stratum and retained-event alphabet controls, but it is strongly mode-dependent under the chosen action. Under this representation, within-fiber inversion distinctions account for 29.9 to 30.1% of held-out reversal score in Bach minor, about 0.1% in Bach major, and 2.9 to 11.6% across WJazzD mode strata; these are corpus-conditional descriptive results, not a genre effect.

10 Directed edges, named windows and stationary cycle circulation

The exploratory secondary-dominant cycle $II^7 \to V^7 \to I^{maj7}$ has the largest edge-product affinity in WJazzD but is supported by one complete forward window in one composition, whereas the

Table 3: Alphabet matching. In each training fold the state alphabet is restricted to the highest-mass complete inversion fibers up to target size k (a fiber is never split; test counts are not used); edges with an endpoint outside the alphabet are dropped from training and test. Achieved size, fixed (self-inverse) and paired fibers, retained test-event mass, groups with no retained event, event-weighted residual $\bar{\delta}_{\text{inv}}$ and its share of $\bar{s}_{C_{12}}$. The final column is the equal-group mean residual minus 1.645 standard errors, in bits, giving a one-sided 95% lower bound; it is neither event-weighted nor a lower bound on the share.

corpus	α	target k	achieved	fixed	paired	retained mass (%)	empty groups	$\bar{\delta}_{\text{inv}}$	share (%)	group LB
Bach	0.1	25	26	0	13	64.6	0	0.162	26.0	0.152
Bach	0.1	50	49	1	24	95.8	0	0.165	17.2	0.155
Bach	0.1	67	68	6	31	99.7	0	0.157	16.3	0.149
WJazzD	0.1	25	24	0	12	46.3	22	0.163	5.3	0.130
WJazzD	0.1	50	53	1	26	67.2	10	0.199	5.8	0.138
WJazzD	0.1	67	74	2	36	75.0	7	0.177	5.0	0.113
Bach	0.5	25	26	0	13	64.6	0	0.154	25.4	0.145
Bach	0.5	50	49	1	24	95.8	0	0.150	16.7	0.142
Bach	0.5	67	68	6	31	99.7	0	0.143	16.0	0.137
WJazzD	0.5	25	24	0	12	46.3	22	0.128	4.4	0.097
WJazzD	0.5	50	53	1	26	67.2	10	0.147	4.8	0.100
WJazzD	0.5	67	74	2	36	75.0	7	0.131	4.2	0.082
Bach	1	25	26	0	13	64.6	0	0.149	25.0	0.140
Bach	1	50	49	1	24	95.8	0	0.143	16.6	0.136
Bach	1	67	68	6	31	99.7	0	0.137	16.0	0.131
WJazzD	1	25	24	0	12	46.3	22	0.111	3.9	0.083
WJazzD	1	50	53	1	26	67.2	10	0.125	4.3	0.084
WJazzD	1	67	74	2	36	75.0	7	0.112	3.9	0.069

conventional ii^7 cycle occurs 54 times in 17 compositions and is never observed in reverse (Table 6). In this spanning-forest basis Bach's stationary circulation is more concentrated than WJazzD's among the five largest terms (top-5 shares in Table 7); these percentages are basis dependent and do not mean that five named progressions explain those fractions of corpus-level asymmetry.

11 Limitations and replication

These results describe two annotated corpora under fixed event definitions, chord templates, key estimates (machine-estimated for Bach, annotated for WJazzD), smoothing rules, and a mode-conditioned inversion that leaves the binary mode covariate fixed. Because the quotient fibers and κ-support are induced by the selected action, another musically defensible inversion action may change both the support gate and the allocation; no action-sensitivity analysis is reported. The held-out quantities are predictive log scores, not physical entropy-production estimates. State alphabets and annotation pipelines differ across corpora and no genre effect is identified. The vocabulary standardisation covers only three shared strata and excludes 29 jazz-only strata. The stationary cycle analysis uses $\alpha = 0.5$, which gives positive fitted flow to pseudocount-supported edges; individual fundamental cycles, ranks and shares depend on the spanning forest and smoothing choice. Named-window counts establish corpus occurrence, not functional or causal harmonic syntax. Code, derived counts, fold assignments and the JSON manifest from which every table regenerates are released in an anonymised repository supplied to the editor.

Table 4: Mode composition. Training uses all events; held-out events are split by the annotated mode of the source chord. κ-support = share of held-out edges whose inverted edge has positive training count under the mode-fixed action. Common-mode weights are fixed at 70.24% major and 29.76% minor for every α. The final column is the equal-group mean residual minus 1.645 standard errors, in bits, giving a one-sided 95% lower bound; it is not a lower bound on the share.

corpus	α	held-out events	n	κ-support (%)	$\bar{s}_{C_{12}}$	$\bar{\delta}_{\text{inv}}$	share (%)	group LB
Bach	0.1	major only	14718	6.8	0.795	0.001	0.1	-0.000
Bach	0.1	minor only	11531	84.0	1.183	0.355	30.0	0.335
Bach	0.1	observed mixture	26249	40.7	0.966	0.156	16.2	0.148
Bach	0.1	common-mode weights	–	–	0.911	0.106	11.7	–
WJazzD	0.1	major only	23031	8.3	3.268	0.115	3.5	0.079
WJazzD	0.1	minor only	4460	10.2	2.903	0.336	11.6	0.113
WJazzD	0.1	observed mixture	27491	8.6	3.209	0.151	4.7	0.094
WJazzD	0.1	common-mode weights	–	–	3.159	0.181	5.7	–
Bach	0.5	major only	14718	6.8	0.747	0.001	0.1	0.000
Bach	0.5	minor only	11531	84.0	1.082	0.324	29.9	0.308
Bach	0.5	observed mixture	26249	40.7	0.894	0.143	16.0	0.136
Bach	0.5	common-mode weights	–	–	0.847	0.097	11.5	–
WJazzD	0.5	major only	23031	8.3	2.814	0.087	3.1	0.059
WJazzD	0.5	minor only	4460	10.2	2.206	0.218	9.9	0.071
WJazzD	0.5	observed mixture	27491	8.6	2.715	0.108	4.0	0.067
WJazzD	0.5	common-mode weights	–	–	2.633	0.126	4.8	–
Bach	1	major only	14718	6.8	0.721	0.001	0.1	0.000
Bach	1	minor only	11531	84.0	1.026	0.309	30.1	0.295
Bach	1	observed mixture	26249	40.7	0.855	0.137	16.0	0.131
Bach	1	common-mode weights	–	–	0.812	0.093	11.4	–
WJazzD	1	major only	23031	8.3	2.602	0.076	2.9	0.051
WJazzD	1	minor only	4460	10.2	1.909	0.172	9.0	0.054
WJazzD	1	observed mixture	27491	8.6	2.490	0.091	3.7	0.056
WJazzD	1	common-mode weights	–	–	2.396	0.104	4.3	–

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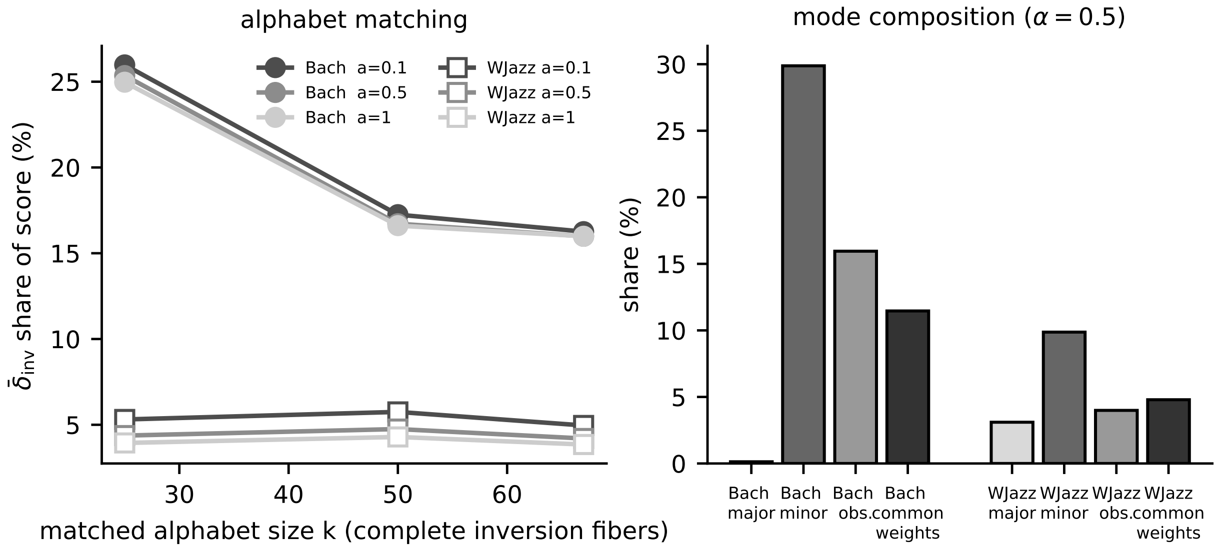

Figure 4: Left: inversion-resolving share under fold-local alphabet matching. Right: share by held-out mode, observed mixture and common-mode weighting ($\alpha = 0.5$).

Table 5: Top directed edges by plug-in contribution to σ (forward and reverse counts).

edge	forward	reverse	bits
Bach			
V maj (major) -> I maj (major)	1611	839	0.058
II dim (minor) -> V maj (minor)	193	4	0.040
V maj (minor) -> I maj (minor)	167	2	0.039
IV maj (major) -> V maj (major)	429	87	0.037
IV min (minor) -> V maj (minor)	214	10	0.035
VII dim (major) -> I maj (major)	390	76	0.035
II maj (major) -> V maj (major)	463	117	0.035
bVII maj (minor) -> bIII maj (minor)	579	206	0.033
WJazzD			
II m7 (major) -> V 7 (major)	1976	107	0.302
V 7 (major) -> I maj7 (major)	984	42	0.162
VI 7 (major) -> II m7 (major)	975	54	0.148
III m7 (major) -> VI 7 (major)	431	5	0.099
V 7 (major) -> I 6 (major)	438	24	0.066
VI 7 (major) -> II 7 (major)	279	3	0.064
I 7 (major) -> IV maj7 (major)	196	0	0.061
II 7 (major) -> V 7 (major)	332	12	0.057

Table 6: Panel A. Named progressions as contiguous windows within works (collapsed events). $N^+ : N^-$ = complete forward and exactly reversed occurrences; the smoothed window ratio is $\log_2 \frac{N^+ + 0.5}{N^- + 0.5}$, not the edge-product affinity A_C. An open path is not a cycle and has no affinity.

corpus	progression	status	$N^+ : N^-$	works with forward	window ratio (bits)
Bach	$i \to iv \to V \to i$	closed	20:0	19	5.358
Bach	$I \to IV \to V \to I$	closed	115:4	94	4.682
Bach	$I \to ii \to V \to I$	closed	165:8	109	4.283
WJazzD	$Imaj7 \to ii7 \to V7 \to Imaj7$	closed	54:0	17	6.768
WJazzD	$Imaj7 \to II7 \to V7 \to Imaj7$	closed	1:0	1	1.585
WJazzD	$Imaj7 \to vi7 \to ii7 \to V7$	open path	0:0	0	0.000

Table 7: Panel B. Spanning-forest circulation terms of the stationary first-order fit ($\alpha = 0.5$; maximum symmetric-traffic forest; return state printed). $\sigma = \sum_C j_C A_C$ closes with error 0.0e+00 (Bach, $\sigma = 0.864$, 2,145 cycles) and 0.0e+00 (WJazzD, $\sigma = 2.156$, 15,051 cycles). Top-5 shares 28.2% and 15.6%, top-10 shares 45.4% and 25.8%. Terms, ranks and shares are basis dependent; only the total is invariant. Stationary σ is a model quantity and is not comparable with the held-out scores of Table 1.

corpus	fundamental cycle	j_C	A_C (bits)	$j_C A_C$	share (
Bach	$Vmajm \to IIdimm \to Iminm \to Vmajm \to Vmajm$	-0.00642	-8.51	0.0546	
Bach	$Vmaj \to IVmaj \to Imaj \to Vmaj \to Vmaj$	-0.01214	-4.37	0.0530	
Bach	$Vmajm \to IVminm \to bIIImajm \to Iminm \to Vmajm \to Vmajm$	-0.00735	-6.49	0.0477	
Bach	$VIIdim \to IVmaj \to Imaj \to VIIdim \to VIIdim$	-0.00642	-7.33	0.0471	
Bach	$Vmaj \to IImin \to Imaj \to Vmaj \to Vmaj$	-0.01098	-3.75	0.0412	
WJazzD	$IIm7 \to I6 \to V7 \to IIm7 \to IIm7$	-0.00604	-13.50	0.0815	
WJazzD	$II7 \to I6 \to V7 \to II7 \to II7$	-0.00576	-13.37	0.0770	
WJazzD	$Imaj7 \to VI7 \to IIm7 \to V7 \to Imaj7 \to Imaj7$	0.00400	16.20	0.0647	
WJazzD	$I7 \to IIIm7 \to VI7 \to IIm7 \to V7 \to I7 \to I7$	0.00249	24.08	0.0600	
WJazzD	$Imaj \to VI7 \to IIm7 \to V7 \to Imaj \to Imaj$	0.00283	18.69	0.0530	

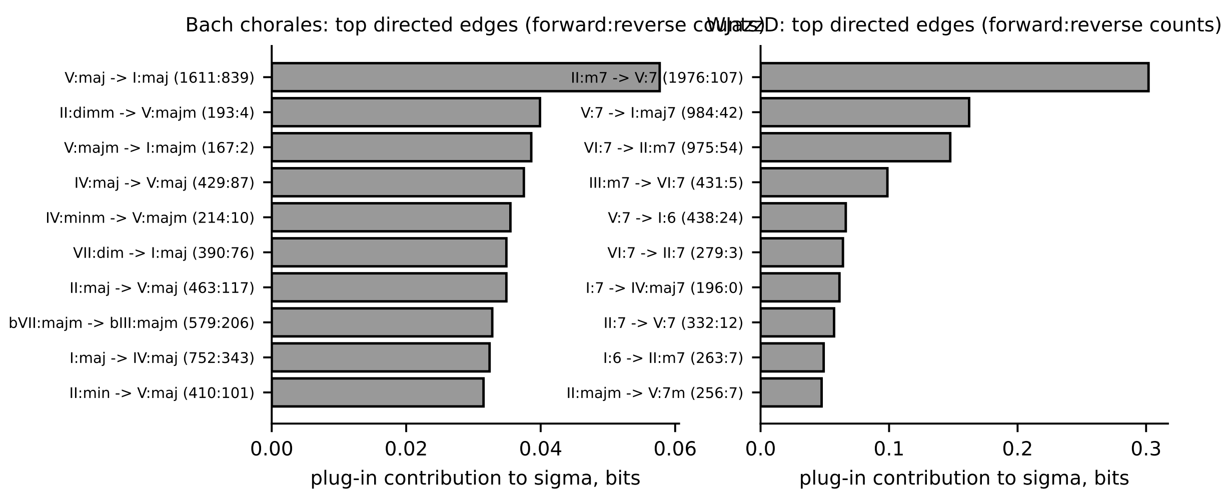

Figure 5: Top directed edges by contribution.

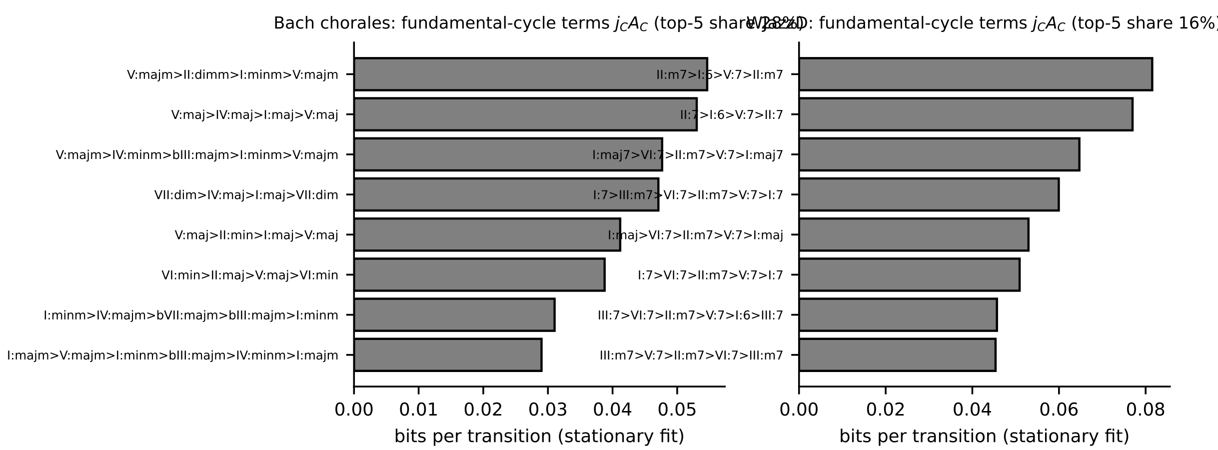

Figure 6: Fundamental-cycle terms of the stationary fit.